

Hu Yang (胡杨)

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Research Interest

My research interest of mathematics lies in the geometry with a categorical fashion, especially in the derived algebraic geometry and geometric representation theory.

Specific topics include higher algebra, higher category theory, $(\infty, 2)$ -category, and their applications.

I am also interested in AI for Math, specific topics include formal theorem proving and AI agent for mathematics.

Education

Sun Yat-Sen university <i>Bachelor in mathematics</i> ◦ GPA: 4.2/5.0, Ranking: 6/86	2020.9 – 2024.6 <i>Guangzhou, China</i>
University of California, Berkeley <i>visiting student at UC Berkeley (BGA)</i>	2023.8 – 2023.12 <i>Berkeley, California, USA</i>
The Hong Kong University of Science and Technology <i>Master of philosophy in mathematics, research-oriented</i> ◦ Supervisor: Quoc Ho, Thesis: Construction of Higher Sheaf theory	2024.8 – 2026.8 <i>Hong Kong SAR</i>
University of Texas, Austin <i>Master of science in computer science</i> Courses taken include: Machine Learning, Deep Learning (implementing deep networks including Transformer and CNN for detector), and Quantum Information	2024.8 – 2027.8 (expected) <i>Online master program</i>

Research Experience

HKUST Summer Research Program <i>HKUST</i> • learning higher category theory and its applications in derived algebraic geometry, especially in a categorified version of Atiyah-Bott formula.	2023.7-2023.8 <i>Tutor: professor Ho Quoc, HKUST</i>
AI for Mathematics <i>Peking University</i> • Learning Lean 4 for formalizing proof in mathematics • I work with a team to formalize the classification of lower dimensional Lie algebras. Here is the repository of our team work.	2024.1
Bayesian protocol for AI agent systems <i>SPAR AI safty</i> • • I work with a team under the lead of professor Aydin Mohseni, focusing on how to build a protocol for allocation of computing power in the Bayesian way. This is an ongoing project.	2026.9-now <i>Tutor: professor Aydin Mohseni, CMU</i>

$(\infty, 2)$ Six Functor Formalism

2024.10-2025.6

HKUST

Tutor: professor Ho Quoc, HKUST

- I develop an $(\infty, 2)$ six functor formalism theory, which encodes the higher sheaf theory developed by German Stefanich.
- The constructed $(\infty, 2)$ six functor formalism can also be used to construct certain braided monoidal structure, for example, on the category of Soergel bimodules.
- This is a joint project with Quoc Ho, Walker H. Stern, and Jonte Gödicke.
- Here is the pdf link of my work

Project Experience

Deep Learning Project

2025.7-2025.8

The University of Texas at Austin

- Coursework-based experience in implementing deep and residual MLP classifiers in PyTorch; developing training and validation pipelines with cross-entropy loss, minibatch optimization, and TensorBoard logging; work on transformer- and CNN-based waypoint prediction for simulated driving.

Workshops and Conferences

Short Course on ‘Mordell-Weil Theorem’

instructor: Liu Qing, University of Bordeaux

Xiamen University

2023.6

Workshop on Complex Geometry and Algebraic Geometry

Tianyuan Mathematics Center in Central China

2023.8

International Workshop on Higher Teichmüller Theory

Chern Institute of Mathematics, Nankai University

2024.6

Higher Structure: Recent Development and its Applications

University of Hamburg

2025.9

Beijing-Shanghai Summer School in Mathematics

Shanghai Institute for Mathematics and Interdisciplinary Sciences

2026.6

Iliad Intensive

Iliad AI safety, Berkeley, USA

2026.10

Specialized Skills

Languages: English (IELTS 7.0)

Programming: MATLAB, Python, C++, Lean

Other Interests

philosophy and psychoanalysis:

I am an amateur in Jacques Lacan’s psychoanalysis. And in the daily life, I am a Platonist, pursuing the ‘Form of the Good’.